Eilik Custom Build — Emotion / Sensor / Movement Datasheet

Hardware assumed:

- Face OLED (emotion states, indexed X/38)
- 6-axis IMU (accelerometer + gyroscope) — body-mounted
- Touch Sensor A — **Chest / Front**
- Touch Sensor B — **Head (top)**
- Touch Sensor C — **Face (screen bezel)**
- 2x arm servos (shoulder swing) ± 1 optional wrist/elbow axis per arm
- 1x head/neck servo (or 2-axis pan-tilt if fitted)
- Body-tilt/base servo (if Eilik base rocks — treat as "body lean")

This is a **design specification**, not a reverse-engineered factory spec — use it as the behavior table to encode in your firmware/animation state machine.

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1. Sensor Trigger Logic (input layer)

Sensor	Physical Meaning	Typical Emotional Response Category
Chest touch (A)	"Pet me / hug me"	Affection, comfort, sleepy states
Head touch (B)	"Pat on the head"	Playful, silly, mischievous, embarrassed
Face touch (C)	"Poke/boop the screen"	Startled, annoyed, questioning, curious
IMU — sudden accel spike	Shake / bump / pickup	Scared, dizzy/confused
IMU — sustained tilt (>30°)	Being tipped over	Scared → Annoyed if repeated
IMU — free-fall (near-zero g)	Dropped	Scared (max intensity)
IMU — long stillness	No motion, no touch for N min	Bored → Tired → Sleepy → Sleeping
Double-tap any sensor within 1s	"excited attention"	Loving, Playful
Long-press (>2s) chest	Deliberate comfort press	Loving

Sensor	Physical Meaning	Typical Emotional Response Category
Rapid multi-touch (any 2 sensors <0.5s apart)	Startle combo	Flushed / Embarrassed

2. Emotion Datasheet (indexed 1–38, full set)

1/38 — SMILE

- **Trigger:** Default/rest state — no active trigger, this is the baseline "happy idle" face returned to after any animation completes
- **Arms:** Both arms at neutral resting position, occasional slow idle sway (breathing-like, 0.3Hz, tiny amplitude)
- **Head:** Centered, neutral, small idle micro-nod every 4–5s
- **Body:** Neutral upright posture
- **LED accent:** Cyan, steady, soft glow

2/38 — GIDDY

- **Trigger:** Quick single tap on chest, or two light touches close together (not full double-tap combo)
- **Arms:** Both arms do a quick small bounce up-down (single bounce, not looping)
- **Head:** Quick tilt-and-return (10°, fast)
- **Body:** Small bounce, single cycle
- **LED accent:** Cyan eyes + pink cheeks, brief brightness pop on trigger

3/38 — ECSTATIC

- **Trigger:** Long-press combo — chest touch held while head is also touched (both sensors together, held 1s+)
- **Arms:** Both arms shoot straight up (full "cheer" pose), small victory shake at top
- **Head:** Tilts back slightly, stays raised
- **Body:** Bounces upward if base allows, or forward lean-and-hold
- **LED accent:** Cyan eyes + pink cheeks, bright pulse, sustained

4/38 — CHARMED

- **Trigger:** Slow, gentle chest touch (light single stroke rather than tap — differentiate by

contact duration/pressure if your sensor supports it)

- **Arms:** Both arms drift up and inward slowly to a soft "aww" pose near chest, held
- **Head:** Gentle tilt to one side, slow
- **Body:** Slight forward lean, relaxed
- **LED accent:** Heart-eyes (white), pink nose/mouth dot, slow warm glow

5/38 — SPARKLING JOY

- **Trigger:** Double-tap chest followed immediately by a head touch (a celebratory combo — e.g., "good job" pat sequence)
- **Arms:** Both arms sweep outward and up in a wide "ta-da" gesture, small sparkle-timed wrist flick at the top
- **Head:** Tilts back slightly, held
- **Body:** Slight rise/bounce
- **LED accent:** Cyan closed-eye arcs with sparkle particles, pink cheeks, brightness flourish

6/38 — PLAYFUL WINK

- **Trigger:** Single quick tap on head only (light pat, <0.3s)
- **Arms:** One arm raises in a quick wave (matching the winking side), other stays neutral
- **Head:** Slight tilt toward the open eye side
- **Body:** Small quick bounce
- **LED accent:** One eye open (cyan circle), one closed (line), yellow smile

7/38 — *(not captured in provided screenshots)*

- This slot in your emotion set wasn't included in the screenshots you shared. If you can grab that one image, I'll fill in its trigger/movement row to keep the datasheet fully continuous — for now, treat it as a placeholder in your state table so indexing doesn't shift.

8/38 — CHEERFUL LAUGH

- **Trigger:** Rapid alternating taps chest→head→chest (a "tickle" pattern, 3 touches within 1.5s)
- **Arms:** Both arms do a quick alternating shimmy (small fast side-to-side, like laughing shoulders)
- **Head:** Quick small bobbing, fast (3Hz), brief
- **Body:** Light bounce/shake, matches laugh rhythm

- **LED accent:** Cyan closed-arc eyes, pink cheeks, open pink mouth — brightness flickers slightly to sell "laughing"

9/38 — EAGER

- **Trigger:** Face touch immediately followed by chest touch (an "ask then confirm" combo), or approach detection if you add proximity sensing later
- **Arms:** Both arms lean forward and slightly up, fingers/end-effector angled toward the touch source (leaning-in gesture)
- **Head:** Leans forward (10–15°)
- **Body:** Forward lean, weight shifted toward front
- **LED accent:** Cyan closed-arc eyes, pink cheeks, small open pink mouth, steady bright

10/38 — ANTICIPATION

- **Trigger:** Chest touch held briefly (0.5–1s), shorter than the Loving long-press threshold
- **Arms:** Both arms rise slightly and pause mid-motion (a "waiting" hover position, not completing a full gesture)
- **Head:** Slight forward tilt, held still
- **Body:** Static, slightly forward
- **LED accent:** Cyan wide eyes with highlight dot, pink cheeks, small yellow smile

11/38 — POUT

- **Trigger:** Touch withdrawn quickly after initiating (e.g., started a chest touch/hold but released before threshold — an "interrupted affection" event)
- **Arms:** Both arms cross low in front (a small "hmph" gesture), held
- **Head:** Slight downward tilt, turned very slightly away
- **Body:** Static, minor recoil
- **LED accent:** Half-lidded cyan eyes (flat top arcs), small downward mouth curve

12/38 — TEARY EYES

- **Trigger:** Escalation state — fires after repeated Pout (11) triggers without a resolving positive touch, or after a Disappointed (14) event persists
- **Arms:** Both arms droop, one arm slowly rises to "wipe" near the eye area
- **Head:** Tilts down slightly
- **Body:** Slight slump
- **LED accent:** Cyan eyes with single tear-drop highlight animation, downward mouth arc

13/38 — CRYING SOFTLY

- **Trigger:** Escalation from Teary Eyes (12) with continued neglect/no comforting touch, or a hard/rough touch event (distinguish via IMU shock + touch combo)
- **Arms:** Both arms rise to cover the face partially, shoulders drawn in
- **Head:** Tilted down, small trembling micro-shake (slow, low amplitude)
- **Body:** Slight forward hunch
- **LED accent:** Falling animated tear-drops under both eyes, small downturned mouth, dimmer overall brightness

14/38 — DISAPPOINTED

- **Trigger:** A touch pattern that "fails" an expected sequence — e.g., robot anticipated a full combo (per Anticipation state) but it didn't complete, or a single flat/unenthusiastic tap after Eager state
- **Arms:** Both arms drop suddenly from wherever they were to a low resting position
- **Head:** Tilts down and slightly to one side
- **Body:** Slight slump, weight back
- **LED accent:** Flat-line eyes (one with small tear dot), small downward mouth arc, dimmed

15/38 — HEARTBROKEN

- **Trigger:** Highest-escalation negative state — e.g., a hard bump (IMU shock) combined with no touch/comfort for an extended window afterward, or repeated Disappointed triggers stacking
- **Arms:** Both arms fall fully to sides and go limp (lowest energy negative pose), slow single droop motion
- **Head:** Fully down, held low
- **Body:** Full slump, lowest posture
- **LED accent:** Broken-heart icon replaces eyes entirely, white glow, no mouth shown, dim overall

16/38 — GRUMPY

- **Trigger:** Face touch while in an already-negative state (Disappointed/Pout), i.e., being poked while already annoyed
- **Arms:** Both arms cross firmly at chest level, held rigid
- **Head:** Slight downward tilt, no movement (stubborn stillness)

- **Body:** Static, rigid
- **LED accent:** Yellow half-lidded eyes (heavy upper lid), small flat downward mouth arc

17/38 — FURIOUS

- **Trigger:** Rapid repeated touches (4+ in 2s) on any single sensor — being repeatedly poked/annoyed past threshold
- **Arms:** Both arms raise stiffly and vibrate/shake in place (barely-contained "shaking with anger")
- **Head:** Very slight forward jut, minimal movement (rigid)
- **Body:** Rigid, small fast tremor
- **LED accent:** Yellow angled "angry brow" eyes, small red mouth arc, brighter than Grumpy

18/38 — FUMING

- **Trigger:** Peak-anger state — escalation from Furious (17) if negative touches continue, or a hard IMU shake combined with prior Furious state
- **Arms:** Both arms raise and cross overhead briefly (a big "steam venting" gesture), quick and sharp
- **Head:** Slight forward tilt, held tense
- **Body:** Small rapid tremor, more intense than Furious
- **LED accent:** Red angled brow-eyes, red mouth/nose dot, brightness flashes (like venting steam)

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3. Emotion Datasheet (indexed 19–38 as captured)

19/38 — ANNOYED

- **Trigger:** Repeated face touch (3+ taps in 2s), or IMU tilt-and-release (being nudged)
- **Arms:** Both arms snap up to a stiff 45° "hands-on-hips" pose, held rigid (no sway)
- **Head:** Sharp single turn away from the touch source (10–15°), held 1s
- **Body:** No lean; a small rigid micro-shake (2Hz, 3° amplitude) for 300ms
- **LED accent:** Yellow (matches eye color)

20/38 — FLUSHED

- **Trigger:** Rapid multi-touch combo (chest+face within 0.5s), or sudden compliment-type interaction pattern

- **Arms:** Both arms rise quickly to cover lower face/cheek area, then retreat halfway (bashful half-cover)
- **Head:** Slight downward tilt (10°), quick left-right micro-wobble
- **Body:** Small backward lean (5°)
- **LED accent:** Pink cheeks pulse (fade in/out, 1Hz)

21/38 — QUESTIONING

- **Trigger:** Single face touch, held (long-press 1s)
- **Arms:** One arm raises to shoulder height with a slight rotation (palm-up "what?" gesture)
- **Head:** Tilt 15° to one side, hold
- **Body:** Static
- **LED accent:** Cyan, steady

22/38 — INVESTIGATING

- **Trigger:** IMU detects being carried/moved slowly (low-magnitude sustained acceleration) or a new sound/light change if you have that sensor
- **Arms:** One arm extends forward/out as if "pointing," other stays at rest
- **Head:** Slow pan sweep left→right (30° arc, 2s duration)
- **Body:** Slight forward lean (5°)
- **LED accent:** Cyan magnifier icon animation on screen; body LEDs steady

23/38 — THOUGHTFUL

- **Trigger:** No touch for 5–10s after an active state (idle decay from Questioning/Investigating)
- **Arms:** One arm slowly rises to "chin" position (resting near face), held
- **Head:** Slow downward tilt (10°), very slow micro-nod every 3s
- **Body:** Static
- **LED accent:** Dim cyan, slow breathing fade

24/38 — CONFUSED SPIRAL

- **Trigger:** IMU shake event (rapid multi-axis acceleration, but below "drop" threshold)
- **Arms:** Both arms spin/rotate at shoulder joint in alternating small circles
- **Head:** Rapid side-to-side wobble (fast, small amplitude, 3Hz)
- **Body:** Full body slow spin-in-place if base servo present; otherwise a wobble/rock
- **LED accent:** Spiral animation synced to green/blue pulse

25/38 — WONDER

- **Trigger:** Head touch followed by stillness (pat, then let go)
- **Arms:** Both arms rise slowly outward and up, palms open, held (an "amazed" gesture)
- **Head:** Slow tilt back (10–15°)
- **Body:** Slight backward lean
- **LED accent:** Cyan with sparkle/twinkle overlay in eyes

26/38 — PLAYFUL

- **Trigger:** Quick double-tap on head or chest
- **Arms:** Asymmetric — one arm waves side to side, other stays down (winking-arm effect matching the winking eye)
- **Head:** Small bob up-down (bouncy, 2Hz)
- **Body:** Light bounce (if base allows vertical give) or forward-back rock
- **LED accent:** Teal + yellow smile mix

27/38 — MISCHIEVOUS

- **Trigger:** Head touch + face touch combo within 1s (sneaky "poke and pat")
- **Arms:** Both arms rise and rub together in front (a "scheming rub-hands" gesture)
- **Head:** Slight forward-down tilt, held with slow side-eye head swivel (small 5° pan)
- **Body:** Slight crouch/lean forward
- **LED accent:** Dim teal, narrowed "eyelash" eye animation

28/38 — CHEEKY TONGUE

- **Trigger:** Single quick face touch (fast tap, <0.3s)
- **Arms:** One arm flicks up briefly in a peace-sign / cheeky wave, then drops
- **Head:** Quick tilt to one side, snap back
- **Body:** Small quick bounce
- **LED accent:** Pink tongue-dot flash

29/38 — SILLY DERP

- **Trigger:** IMU detects being spun or tilted rapidly then released (dizzy-inducing motion) OR triple-tap any sensor
- **Arms:** Both arms flop loosely side to side asynchronously (limp, floppy motion — lower servo stiffness/torque temporarily if adjustable)
- **Head:** Loose wobble, asymmetric (uneven left/right swing)

- **Body:** Slight stagger side to side
- **LED accent:** Mismatched pupil-size eyes (as shown), teal

30/38 — LOVING

- **Trigger:** Long-press chest touch (>2s) — the "hug" gesture
- **Arms:** Both arms sweep inward to a "self-hug" pose across the chest, held
- **Head:** Gentle forward nod-tilt (10°), slow
- **Body:** Slow forward lean, then settle
- **LED accent:** Pink heart-eyes, warm glow pulse (slow, 0.5Hz — like a heartbeat)

31/38 — TIRED

- **Trigger:** IMU stillness timer stage 1 (e.g., no motion/touch for 2–5 min)
- **Arms:** Both arms droop slowly to sides, slight sag at "elbow" if articulated
- **Head:** Slow downward tilt (15–20°), stays low
- **Body:** Slight overall slump (lower torso servo height if you have one, or just tilt)
- **LED accent:** Dim cyan, half-closed eyes, slow fade

32/38 — SLEEPING ZZZ

- **Trigger:** IMU stillness timer stage 3 (extended inactivity, e.g., 10+ min) — deepest idle state
- **Arms:** Both arms fully relaxed at sides, motors can de-energize/idle (torque off) to save power and look "asleep"
- **Head:** Fully drooped, minimal micro-movement (simulate breathing: 0.2Hz tiny nod)
- **Body:** Fully settled/lowest position
- **LED accent:** Very dim, "Z Z" text floats/fades on screen, eyes closed (flat lines)
- **Wake condition:** Any touch on any sensor, or IMU motion above wake-threshold → transitions to 34 (Sleepy) as the wake-up intermediate, not straight to alert

33/38 — YAWNING

- **Trigger:** Transitional animation, auto-fired when moving from Tired (31) into Sleeping (32), or on wake from Sleeping before reaching Sleepy
- **Arms:** Both arms slowly raise into a "stretch" pose (up and out), hold 1s, then lower
- **Head:** Tilts back briefly during the "yawn," then returns
- **Body:** Slight backward lean during stretch
- **LED accent:** Orange/yellow mouth-dot, eyes closed arcs

34/38 — SLEEPY

- **Trigger:** Wake-transition state (touch/motion while in Sleeping), or IMU stillness timer stage 2
- **Arms:** Slow, heavy-looking movement — arms lift partway then settle back down (struggling to stay "awake")
- **Head:** Nodding drift (slow downward drift, then a corrective jerk back up every 2–3s — the classic "nodding off" motion)
- **Body:** Minor sway, low energy
- **LED accent:** Dim cyan, half-arc eyes, small dot mouth

35/38 — SCARED

- **Trigger:** IMU sudden high-g spike (shake/bump/near-drop), or unexpected simultaneous touch on all 3 sensors
- **Arms:** Both arms snap up and inward defensively (guard/flinch pose), fast movement, held rigid
- **Head:** Snaps backward away from stimulus, quick
- **Body:** Rapid backward lean/recoil, then a settling micro-tremor (fast small shake, decaying amplitude — like a startle recovery)
- **LED accent:** Bright cyan wide "pupil" dot eyes, high brightness flash on trigger

36/38 — BORED MEH

- **Trigger:** No interaction for a moderate period while robot is "awake" (shorter timeout than Tired — an intermediate disengagement state), or repeated same touch input with no variation
- **Arms:** Both arms hang loosely, occasional slow single-arm shrug (one shoulder lift-drop) every 5–8s
- **Head:** Slight side droop, minimal movement
- **Body:** Static, low energy
- **LED accent:** Dim teal, half-lidded flat eyes

37/38 — EMBARRASSED

- **Trigger:** A "failed" or interrupted gesture — e.g., robot was mid-animation and got touched/bumped unexpectedly, or IMU detects near-tip (partial tilt then recovered)
- **Arms:** One arm rises to cover part of the face, other drops awkwardly
- **Head:** Tilts down and away, small fast side-shake (like shaking off embarrassment)
- **Body:** Slight crouch/shrink pose

- **LED accent:** Pink cheek blush, small closed-line mouth

38/38 — HUNGRY

- **Trigger:** Timer-based (e.g., a "feeding" mechanic in your firmware — X minutes since last "feed" interaction), or a dedicated feed gesture (e.g., swipe on face sensor)
- **Arms:** Both arms reach forward and slightly up repeatedly (a "gimme" reaching motion), looping every 1.5s
- **Head:** Tracks/follows toward chest sensor (as if looking at food source), slight forward lean
- **Body:** Rocks forward-back gently, anticipatory
- **LED accent:** Green eyes with the fish/star icon animation, mouth small oval

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4. State-Machine Notes

Priority order (highest wins if multiple triggers fire at once):

1. IMU drop/fall detection → Scared (35) — safety-critical, always overrides
2. Any touch input → corresponding emotional response
3. IMU shake/tilt → Scared / Confused Spiral
4. Idle timers → Bored → Tired → Sleepy → Sleeping (linear decay chain)
5. Feed timer → Hungry (can interrupt idle decay chain)

Suggested idle decay chain (no interaction): `Bored (36) → Tired (31) → Yawning (33) → Sleeping (32)`

Suggested wake chain (touch/motion during sleep): `Sleeping (32) → Yawning (33) → Sleepy (34) → [normal active state]`

Suggested "sad" escalation chain (neglect/rough handling): `Pout (11) → Teary Eyes (12) → Crying Softly (13) → Disappointed (14) → Heartbroken (15)` A comforting touch (long-press chest) at any point in this chain should reset toward Loving (30) or Smile (1) rather than requiring the full chain to play out.

Suggested "anger" escalation chain (repeated poking/rough touch): `Grumpy (16) → Furious (17) → Fuming (18)` This should decay back to Annoyed (19) or Smile (1) after a cooldown period with no further negative touches, rather than staying pinned at Fuming.

Suggested "positive" escalation chain (good interaction streaks): `Giddy (2) → Ecstatic (3) → Sparkling Joy (5)`, with Charmed (4), Eager (9), Anticipation (10), and Cheerful Laugh (8) as side-branches depending on which sensor combo fires.

Servo torque tip: For "floppy" states (Silly Derp 29, Sleeping 32, Tired 31), consider temporarily lowering PWM/torque limits on arm servos rather than just changing target angles — it reads as far more convincing than a stiff servo moving to a "droopy" angle.

Debounce recommendation: Touch sensors should debounce at ~150–200ms and use a rolling timestamp buffer to detect "combo" touches (e.g., Mischievous = head+face within 1s) reliably without false triggers from a single accidental touch.

This datasheet is a design template based on your 20 captured emotion states — tune all timing values (Xs thresholds, decay chain durations) to match your servo speed and IMU sensitivity during bench testing.